Table 1: RN-CatBoost real-data diagnostic from the reliable-negative mechanism audit. Ranking differences are RN minus naive CatBoost; each dataset/design value averages 20 splits within each budget and then the two budgets. Selector diagnostics refer to the final train+validation RN set.

One-sample					
Dataset	AUROC pp	AP pp	Purity %	Neg. recall %	Selected U %
Credit Default	-4.228	-5.166	91.84	11.59	10.15
Dry Bean	+0.152	+0.958	98.37	93.32	72.43
HTRU2	+0.141	+2.054	99.13	82.07	76.09
Online Shoppers	-0.815	-0.251	95.19	20.99	19.02
Optical Digits	-0.793	-0.933	92.17	53.57	31.36
Phishing Websites	+1.263	+1.630	94.01	69.73	38.58
Rice	+0.889	+2.199	94.74	87.53	56.06
Spambase	+1.416	+2.265	94.24	63.26	42.74
Two-sample					
Dataset	AUROC pp	AP pp	Purity %	Neg. recall %	Selected U %
Credit Default	-4.481	-5.906	90.98	9.29	7.98
Dry Bean	+0.369	+1.984	98.71	90.90	67.96
HTRU2	+0.078	+1.836	99.02	82.70	75.92
Online Shoppers	-0.891	+0.174	94.56	24.54	22.10
Optical Digits	-0.555	-0.782	91.94	57.15	31.66
Phishing Websites	+1.309	+1.758	93.76	66.64	34.42
Rice	+0.932	+2.161	93.89	86.40	52.99
Spambase	+1.345	+2.115	92.83	61.45	40.06

Purity is the true-negative fraction among selected U; negative recall is retained true negatives divided by available true negatives; selected U is the fraction of U retained. RN sets were previously reconstructed from exact archived inputs/code because original indices were unavailable. No selectors are rerun here. These associations do not show that purity causes a performance difference; the selector sensitivity analysis alternative selectors are not substituted for original RN.